

YASHRAJ RASTOGI

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PROFESSIONAL SUMMARY

B.Tech Computer Science undergraduate (2024–2028) specializing in full-stack web development, AI integration, and scalable system architecture. Recognized as a Top 2% National Finalist (40,000+ participants) at the India AI Impact Buildathon and Top 10 Finalist at CodeRush Hackathon. Proficient in Next.js, React, Firebase, and PyTorch with hands-on experience building real-time, AI-driven applications using Google Gemini. Demonstrated leadership as a Google Student Ambassador (GID: 3263) and Technical Head.

TECHNICAL SKILLS

Languages:	Java, JavaScript, TypeScript, Python, C
Frontend:	React.js (v18/v19), Next.js, Tailwind CSS v4, HTML5, Progressive Web Apps (PWA)
Backend:	Node.js, Express.js, Django, FastAPI, Cloud Functions (Serverless)
AI / ML:	Google Gemini AI, PyTorch, Hugging Face, Wav2Vec 2.0, Librosa
DB & Cloud:	MongoDB, Firebase (Firestore, Auth, Rules), Vercel, Hugging Face Spaces
Tools:	Git, GitHub, Figma, Canva, Vite, Postman
Concepts:	Full-Stack Development, REST API Design, System Architecture, Explainable AI (XAI), Role-Based Access Control

PROJECTS

Crisis OS — Real-Time Emergency Management PWA

2026

React 18 · TypeScript · Firebase · Google Gemini AI · PWA · Tailwind CSS | GDG Solution Challenge 2026 — Team Leader

[Live Demo](#) · [GitHub](#)

- Architected a real-time PWA supporting five distinct user roles (Admin, Manager, Staff, Guest, Responder) with role-based architecture capable of handling intense concurrent operational load.
- Built a live situational dashboard using Firebase Cloud Firestore real-time listeners, enabling managers to track guest safety status (Safe / Need Help / Critical) by floor and zone without manual refresh.
- Integrated Google Gemini AI into the reporting pipeline to automatically convert raw field reports into severity-ranked action checklists and targeted broadcast drafts.
- Designed a secure serverless backend with Node.js Cloud Functions for automated role assignments and strict Firestore Security Rules ensuring complete data isolation across properties.
- Implemented protected routing and state management for secure QR-code-based guest onboarding and isolated emergency handoff links for responders.

UniVerse — Real-Time Social Platform

2026

Next.js · TypeScript · Gemini AI 1.5 Flash · Firebase · Tailwind CSS v4 · Shadcn/UI | Top 10 Finalist — CodeRush Hackathon

[Live Demo](#) · [GitHub](#)

- Developed a full-stack Next.js application using the App Router and React 19, implementing scalable architecture for a real-time social platform with themed interaction zones.
- Built an automated content moderation pipeline using Gemini AI 1.5 Flash that dynamically analyzes and filters inappropriate content in anonymous user posts in real time.
- Implemented Firebase Auth and Cloud Firestore for real-time data synchronization and secure role-based access across the platform.
- Designed a multi-tiered UI with Tailwind CSS v4 and Shadcn/UI, featuring dynamic global dark mode and a distinct contextual theme for anonymous spaces.
- Managed end-to-end deployment via Vercel, optimizing build configurations for rapid prototyping and continuous iteration.

VoiceGuard — AI Voice Detection REST API

2026

Python · FastAPI · PyTorch · Hugging Face · Wav2Vec 2.0 (XLSR-53) | Top 2% — India AI Impact Buildathon (40,000+ participants) [Live Demo](#) · [GitHub](#)

- Engineered a hybrid AI detection pipeline combining a fine-tuned Wav2Vec 2.0 (XLSR-53) transformer with audio forensic algorithms to identify synthetic voice anomalies and acoustic artifacts.

- Built a high-performance REST API using FastAPI and Uvicorn capable of real-time, low-latency audio analysis via Base64-encoded payload submissions.
- Implemented Explainable AI (XAI) features parsing diagnostic model metrics (e.g., artificial spectral smoothness) into human-readable summaries for non-technical reviewers.
- Containerized and deployed the application as a live public demo on Hugging Face Spaces with a standalone web interface for seamless file testing.

AcadVault — Academic Management Portal

2025

React.js · Tailwind CSS · Firebase | Smart India Hackathon (SIH 2025) — Internal Round Qualifier [Live Demo](#) · [GitHub](#)

- Built a comprehensive academic management portal for students, faculty, and administrators, qualifying through internal SIH collegiate rounds.
- Implemented role-based access control enabling students to manage profiles, faculty to verify achievements, and administrators to update academic records and generate portfolios.
- Developed an admin dashboard for user management and automated data aggregation to streamline NAAC and NIRF accreditation report generation.

Heat Sense — Hotel Staff Fire Safety Trainer

2025

React.js · Firebase · JavaScript · HTML/CSS | GDG Solution Challenge 2025 — Participant

- Developed a mobile-first web application using scenario-based quizzes to train hotel staff for fire emergencies, built for the GDG on Campus Solution Challenge 2025.
- Upgraded initial prototype to a full-stack React application, integrating Firebase for backend infrastructure, authentication, and role-based user journeys.
- Implemented secure authentication enabling administrators to onboard properties and manage staff profiles seamlessly.

LEADERSHIP & EXPERIENCE

Google Student Ambassador · Google India · GID: 3263

May 2026 – Present

- Organize monthly hands-on technical workshops campus-wide advocating for Google AI tools including Gemini and NotebookLM, managing end-to-end event logistics.
- Provide 1-on-1 technical consultations to student founders, identifying operational bottlenecks and integrating tools like NotebookLM and Gemini Canvas to accelerate project development.
- Drive digital engagement through targeted social media content showcasing real-world LLM use cases and fostering a tech-forward builder community on campus.

Technical Head · Unstop Igniters Club SRMCEM

Apr 2026 – Present

- Lead technical teams in website development and digital infrastructure initiatives supporting the club's events and online presence.

Technical Coordinator · AlgoZenith Chapter SRMCEM

Oct 2025 – Present

- Organized the 'Vibe Coding Contest' hackathon and 'Algo-Arena' DSA competition, managing logistics and facilitating competitive programming for students.
- Coordinated technical workshops led by industry professionals to enhance practical skills within the collegiate community.

EDUCATION

B.Tech in Computer Science and Engineering

2024 – 2028

Shri Ramswaroop Memorial College of Engineering & Management, Lucknow · CGPA: 7.5 / 10

Intermediate Science (12th Grade)

2022 – 2023

City Convent School · 90.25% (9.02 CGPA)

High School (10th Grade)

2020 – 2021

City Convent School · 84% (8.4 CGPA)

CERTIFICATIONS & AWARDS

- Top 2% National Finalist — India AI Impact Buildathon, HCL GUVI (40,000+ participants) [Feb 2026] [View Credential](#)
- Top 10 Finalist — CodeRush Hackathon (UniVerse Project) [2026]
- Team Leader — GDG Solution Challenge 2026 (Crisis OS) [Result Pending]
- Team Leader / Participant — GDG Solution Challenge 2025 (Heat Sense) [2025] [View Credential](#)
- HeLa Crossroads Certification — HeLa Labs [2024] [View Credential](#)
- Web Development Certification — Sololearn [2024] [View Credential](#)
- Java Certification — Sololearn [2024] [View Credential](#)